Research Article

Fuzzy-Based Analysis and Performance Evaluation of Classical Queuing Models under Uncertain Arrival and Service Rates

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Abstract

Classical queueing systems assume static arrival and service parameters; however, real-world operational environments exhibit variance due to incomplete data or fluctuating demand. This study models this parameter imprecision by defining arrival and service rates via triangular fuzzy numbers and propagating their α-cut bounds through standard M/M/1 metrics. We evaluate the fuzzy expected system size, queue length, system duration, and waiting time. System stability is verified pointwise across all α-levels, ensuring localized instable regimes are not masked by single aggregated indicators. The analysis is subsequently extended to M/M/c environments by introducing a worst-case utilization robustness index. We establish three main theorems governing α-level stability, index monotonicity, and minimum server bounds for targeted robustness levels. An application to hospital emergency department staffing demonstrates that while four physicians suffice under nominal conditions, five are necessary to maintain a 20\% robustness threshold across the full uncertainty spectrum.

Keywords

Fuzzy queuing system; Queueing theory; Fuzzy numbers; Triangular fuzzy numbers; Arrival rate; Service rate; Mean queue length; Waiting time; Performance evaluation; Uncertainty.

2020 Mathematics Subject Classification

60K25, 60K30, 90B22, 03E72.

1. Introduction

Queuing theory is an important area of applied mathematics and operations research concerned with the mathematical analysis of waiting lines and congestion phenomena. Queueing theory provides the mathematical apparatus for analyzing waiting lines and service bottlenecks across healthcare, telecommunications, transportation, and manufacturing networks. A typical queuing system consists of an arrival process, a waiting line, one or more service facilities, and a service discipline that determines the order in which customers are served. The mathematical description of a queue is commonly expressed through Kendall's notation, in which the arrival process, service-time distribution, and number of servers constitute the basic structural components of the model [1]. Classical queueing theory provides several well-established models, including the (M/M/1), (M/M/c), (M/G/1), (G/M/1), and finite-capacity queueing systems, which allow important operational characteristics to be evaluated analytically [1,2].

The performance of a queuing system is generally assessed through measures such as traffic intensity, probability of system states, expected number of customers in the system, mean queue length, average waiting time, and server utilization. For example, in a stable single-server (M/M/1) system, the traffic intensity is determined by the ratio of the arrival rate to the service rate. Classical queueing models are mathematically tractable because they generally assume that the underlying parameters and probability distributions are known with sufficient precision. Little's law provides a fundamental relationship between the average number of customers in a system and the average time spent in that system, expressed as (L=\lambda W), and remains applicable across a broad class of queueing systems [2, 3]. Such analytical results have made classical queueing theory a powerful tool for system design, capacity planning, and performance evaluation.

Despite their usefulness, conventional queueing models may not adequately represent many real-life service systems in which system parameters cannot be specified precisely. In practical situations, statements such as "approximately 10 customers arrive per hour", "the average service time is around five minutes", or "the service rate is relatively high" may be more realistic than exact numerical specifications. Moreover, arrival and service

processes may be influenced by human behavior, environmental conditions, demand fluctuations, incomplete observations, and expert judgment. Consequently, uncertainty in practical queueing systems may arise not only from randomness but also from vagueness and imprecision in the available information. The synopsis underlying the present

study similarly identifies arrival rate, service rate, waiting time, and queue length as important parameters whose exact values may not be available in practical applications.

Fuzzy set theory provides a natural mathematical framework for dealing with such imprecision. The theory was introduced by Zadeh [4], who generalized the classical notion of set membership by allowing an element to possess a membership grade in the interval [0,1]. Unlike probability theory, which primarily addresses randomness, fuzzy set theory is particularly useful for representing vagueness and imprecise linguistic information. Thus, quantities such as "low arrival rate", "moderate service rate", or "approximately 15 customers per hour" can be represented mathematically through suitable membership functions. This characteristic makes fuzzy methodology particularly relevant to queueing systems where the available information is based on estimates, expert opinions, linguistic descriptions, or incomplete observations.

The application of fuzzy set theory to queueing systems has developed into an important research direction generally referred to as fuzzy queueing theory. Early work by Prade [5] considered fuzzy or possibilistic approaches to queueing systems, providing a foundation for incorporating imprecise information into queueing analysis. Subsequently, Li and Lee [6] proposed a general framework for queueing systems in fuzzy environments using Zadeh's extension principle, possibility concepts, and fuzzy Markov chains. Their work demonstrated analytical treatment of fuzzy queueing systems through models such as (M/F/1) and (FM/FM/1), thereby establishing an important connection between classical queueing theory and fuzzy mathematical modeling.

Negi and Lee [7] further investigated the analysis and simulation of fuzzy queues by combining fuzzy-set representations with conventional probabilistic queueing approaches. Their methodology employed (\alpha)-cut representations and simulation techniques, with trapezoidal fuzzy numbers used to describe uncertain parameters. Their results

demonstrated that fuzzy and probabilistic approaches can be combined to obtain queueing models that are more flexible than conventional crisp models when the input information is imprecise.

An important development in fuzzy queueing analysis was presented by Kao, Li, and Chen [8], who proposed a parametric programming approach for constructing membership functions of queueing performance measures when interarrival and service-time parameters are fuzzy. Their method transforms a fuzzy queue into a family of corresponding crisp queueing systems through the use of (\alpha)-cuts and then derives membership functions for the resulting performance measures. The approach was demonstrated for several single-server fuzzy queueing configurations, including (M/F/1), (F/M/1), (F/F/1), and (FM/FM/1) models.

The fuzzy treatment of queueing systems has subsequently been extended to more complex settings, including finite-capacity systems, unreliable servers, retrial queues, heterogeneous servers, priority queues, and systems with other operational constraints. For example, fuzzy analysis has been used to study queueing systems with unreliable servers through nonlinear programming approaches, while fuzzy parameters have also been incorporated into retrial and heterogeneous-server models [9,10]. These developments indicate that fuzzy queueing theory can accommodate a broad range of operational uncertainties rather than being restricted to simple single-server systems.

Simulation has also played an important role in the development of fuzzy queueing models. Wu [11] investigated simulation procedures for queueing systems under fuzziness and explicitly considered fuzzy arrival and service rates. Simulation-based approaches are particularly useful when analytical derivation of membership functions becomes difficult because of the nonlinear interaction between several uncertain parameters. Thus, analytical fuzzy modeling and computational simulation can be viewed as complementary approaches for investigating the behavior of uncertain queueing systems.

In the present study, the focus is placed on the systematic analysis of classical queueing models when important system parameters are represented by fuzzy numbers. In particular, the arrival rate $\tilde{\lambda}$ and service rate $\tilde{\mu}$ are considered uncertain quantities. A triangular fuzzy number may be used in the form $\tilde{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$, $\tilde{\mu} = (\mu_1, \mu_2, \mu_3)$, where the first and third components represent the lower and upper possible values, while the middle component represents the most plausible value. This representation is especially suitable when precise statistical observations are unavailable but reasonable lower, most likely, and upper estimates can be obtained from historical data or expert knowledge. The uploaded synopsis also proposes triangular fuzzy numbers for uncertain arrival and service rates and suggests incorporating these values into conventional queueing expressions through fuzzy arithmetic and extension principles.

The principal motivation of the present investigation is therefore to examine how uncertainty in fundamental queueing parameters affects the resulting performance characteristics. In a crisp model, a particular pair $(\tilde{\lambda}, \tilde{\mu})$ produces a single value of a performance measure. In contrast, when $\tilde{\lambda}$ and $\tilde{\mu}$ are fuzzy quantities such as traffic intensity, mean queue length, average waiting time, and expected number of customers in the system become fuzzy performance measures. The resulting membership functions preserve information about the uncertainty of the inputs and can provide decision-makers with a more informative description of system behavior than a single deterministic estimate.

The study further aims to compare conventional and fuzzy versions of representative queueing models. Such a comparison is important because the classical model remains useful when accurate numerical information is available, whereas the fuzzy model is expected to be more appropriate when parameters are estimated or linguistically described. The objective is not to replace probability-based queueing theory but to provide an additional modeling framework capable of incorporating vagueness into the analysis. The comparison of crisp and fuzzy models can consequently reveal the extent to which parameter imprecision influences queue length, waiting time, and overall system performance.

Accordingly, this research develops a mathematical framework for analyzing queueing systems under uncertain arrival and service conditions. The study considers the basic structure and components of queueing systems, incorporates fuzzy parameters into appropriate queueing formulations, evaluates important performance measures, and compares the resulting fuzzy models with their classical counterparts. These objectives are consistent with the research objectives stated in the submitted synopsis. The proposed framework is intended to provide a more realistic representation of service systems in which exact parameter values are difficult to determine and to establish a foundation for further development of fuzzy multi-server, finite-capacity, priority, and other generalized queueing models.

2. PRELIMINARIES

In this section, we recall some basic concepts of queueing theory and fuzzy set theory that are required for the subsequent development of the proposed fuzzy queuing models.

2.1 Basic Components of a Queuing System

A queuing system consists essentially of customers arriving for service, a waiting line, one or more service facilities, and a service discipline. The principal parameters used to characterize a queuing system are the arrival rate, service rate, number of servers, queue capacity, and queue discipline.

Let λ denote the mean arrival rate of customers per unit time and let μ denote the mean service rate of a server per unit time. For a single-server system, the traffic intensity is defined by

$$\rho = \lambda / \mu.$$

For a stable M/M/1 queue, the stability condition is

$$0 \le \rho < 1.$$

The principal performance measures include:

- L: average number of customers in the system;
- Lq: average number of customers waiting in the queue;
- W: average time spent by a customer in the system;
- Wq: average waiting time in the queue;
- ρ: utilization or traffic intensity of the server.

For an M/M/1 queue, these measures are given by

$$L = \lambda / (\mu - \lambda),\ Lq = \lambda^2 / [\mu(\mu - \lambda)],$$
$$W = 1 / (\mu - \lambda),\ Wq = \lambda / [\mu(\mu - \lambda)].$$

These relationships provide the classical foundation for constructing their fuzzy counterparts.

2.2 Little's Law

Little's law establishes a fundamental relationship between the average number of customers in a system, the arrival rate, and the average time spent in the system. It is expressed as

$$L = \lambda W.$$

Similarly, for customers waiting in the queue,

$$Lq = \lambda Wq.$$

In the fuzzy environment, when the arrival rate and waiting time are represented by fuzzy quantities, the corresponding relationship may be expressed as

$$\tilde{L} = \tilde{\lambda} \otimes \tilde{W},$$

where $\otimes$ denotes fuzzy multiplication.

2.3 Fuzzy Set

A fuzzy set was introduced by Zadeh [1] to represent imprecise or vague information. Let X be a non-empty universal set. A fuzzy set Ã on X is characterized by a membership function

$$\mu_{\tilde{A}} : X \rightarrow [0,1].$$

Thus,

$$\tilde{A} = \{(x, \mu_\tilde{A}(x)) : x \in X\},$$

where $\mu_{\tilde{A}}(x)$ represents the degree of membership of x in Ã. The value 0 indicates complete non-membership, whereas 1 indicates complete membership.

2.4 Fuzzy Number

A fuzzy set Ã defined on the real line R is called a fuzzy number if its membership function satisfies the following conditions:

1. Ã is normal, i.e., there exists at least one $x \in R$ such that $\mu_{\tilde{A}}(x) = 1$.
2. Ã is convex, i.e., $\mu_{\tilde{A}}(\lambda x + (1 - \lambda)y) \geq \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{A}}(y)\}$ for all $x, y \in R$ and $\lambda \in [0,1]$.
3. The membership function is upper semi-continuous.
4. The support of Ã is bounded.

Fuzzy numbers are useful for representing quantities such as "approximately 10 customers per hour" or "approximately 5 minutes of service time."

2.5 Triangular Fuzzy Number

A triangular fuzzy number is one of the simplest and most widely used fuzzy representations. A triangular fuzzy number is denoted by

$$\tilde{A} = (a_1, a_2, a_3),$$

where $a_1 \leq a_2 \leq a_3$. Its membership function is defined as

$$\mu_{\tilde{A}}(x) = \{ 0, x < a_1;\ (x - a_1)/(a_2 - a_1), a_1 \leq x \leq a_2;\ (a_3 - x)/(a_3 - a_2), a_2 \leq x \leq a_3;\ 0, x > a_3. \}$$

Here, a_1 represents the lower value, a_2 represents the most plausible or modal value, and a_3 represents the upper value.

In the proposed fuzzy queuing framework, the arrival and service rates may be represented as triangular fuzzy numbers:

$$\tilde{\lambda} = (\lambda_1, \lambda_2, \lambda_3),$$

$$\tilde{\mu} = (\mu_1, \mu_2, \mu_3).$$

This representation is particularly useful when exact arrival and service rates are unavailable but lower, most likely, and upper estimates can be obtained.

2.6 α-Cut of a Fuzzy Number

For a fuzzy number Ã, its α-cut for $\alpha \in [0,1]$ is defined by

$$[\tilde{A}]^{\alpha} = \{x \in R : \mu_\tilde{A}(x) \geq \alpha\}.$$

For the triangular fuzzy number $\tilde{A} = (a_1, a_2, a_3)$, the α-cut is

$$[\tilde{A}]^{\alpha} = [a_1 + \alpha(a_2 - a_1), a_3 - \alpha(a_3 - a_2)].$$

The α-cut representation is particularly useful for transforming a fuzzy queuing problem into a family of interval-valued or crisp problems.

2.7 Arithmetic Operations on Triangular Fuzzy Numbers

Let $\tilde{A} = (a_1, a_2, a_3)$ and $\tilde{B} = (b_1, b_2, b_3)$ be two positive triangular fuzzy numbers. Their addition is defined by

$$\tilde{A} \oplus \tilde{B} = (a_1 + b_1, a_2 + b_2, a_3 + b_3).$$

For subtraction, under the usual interval-extension interpretation,

$\tilde{A} \ominus \tilde{B} = (a_1 - b_3, a_2 - b_2, a_3 - b_1)$.

For positive fuzzy numbers, multiplication can be represented by

$\tilde{A} \otimes \tilde{B} = (a_1b_1, a_2b_2, a_3b_3)$.

Similarly, for positive values,

$\tilde{A} / \tilde{B} = (a_1/b_3, a_2/b_2, a_3/b_1)$.

These operations allow classical queueing expressions to be extended to uncertain parameters.

2.8 Fuzzy Arrival and Service Rates

In a classical queueing model, the arrival rate λ and service rate μ are usually treated as precise numerical quantities. Under uncertainty, these parameters can instead be represented by fuzzy numbers:

$$\tilde{\lambda} = (\lambda_1, \lambda_2, \lambda_3), \quad \tilde{\mu} = (\mu_1, \mu_2, \mu_3).$$

The fuzzy traffic intensity is then represented as

$$\tilde{\rho} = \tilde{\lambda} / \tilde{\mu}.$$

For a stable fuzzy single-server queue, the service rate should dominate the arrival rate over the relevant range of possible values. A sufficient conservative condition is

$$\lambda_3 < \mu_1.$$

Under this condition, the queue remains stable throughout the considered uncertainty range.

2.9 Fuzzy Performance Measures

Once the arrival and service rates are represented by fuzzy numbers, the principal performance measures can also be expressed fuzzily. For example, the fuzzy mean number of customers in an M/M/1 system can be represented by

$$\tilde{L} = \tilde{\lambda} / (\tilde{\mu} - \tilde{\lambda}),$$

while the fuzzy mean waiting time in the system is

$$\tilde{W} = 1 / (\tilde{\mu} - \tilde{\lambda}).$$

Similarly,

$\tilde{L}q = \tilde{\lambda}^2 / [\tilde{\mu}(\tilde{\mu} - \tilde{\lambda})]$ $\tilde{W}q = \tilde{\lambda} / [\tilde{\mu}(\tilde{\mu} - \tilde{\lambda})]$.

These expressions provide the mathematical basis for comparing the performance of classical and fuzzy queueing systems.

2.10 Defuzzification

In certain decision-making applications, it may be necessary to convert a fuzzy performance measure into a single representative numerical value. This process is called defuzzification. One commonly used approach is the centroid method. For a fuzzy quantity $\tilde{A}$, the centroid value may be expressed as

$$A^* = [\int x \, \mu_{\tilde{A}}(x) \, dx] / [\int \mu_{\tilde{A}}(x) \, dx].$$

The resulting crisp value can then be compared directly with the corresponding performance measure obtained from a classical queueing model.

The above concepts provide the necessary mathematical foundation for the proposed analysis of fuzzy queueing systems with uncertain arrival and service rates.

3. Fuzzy Queueing Model and Comparative Analysis

This section develops a new, transparent fuzzy queueing framework based on α-cut interval propagation. The model retains uncertainty in fuzzy arrival and service rates while producing direct interval-valued performance measures at prescribed α-levels.

3.1 Motivation and Research Gap

Li and Lee developed an analytical fuzzy-queue framework using Zadeh's extension principle, possibility concepts and fuzzy Markov chains, with examples including M/F/1 and FM/FM/1 systems. Negi and Lee combined fuzzy-set representations with conventional queueing approaches and investigated α-cut and simulation procedures using

trapezoidal fuzzy numbers. Kao, Li and Chen proposed an α-cut and parametric-programming procedure for constructing membership functions of performance measures in several fuzzy one-server models.

Building on these foundations, the present work proposes a compact α-cut interval algorithm for a fuzzy M/M/1 queue. Its distinguishing feature is the explicit stability screening at every α-level, simultaneous calculation of L, Lq, W and Wq, and direct comparison with the classical modal model. This is a new framework developed here for transparent numerical analysis; it is not claimed to reproduce the exact algorithms of the earlier papers.

3.2 Definition

Let the uncertain arrival and service rates be represented by triangular fuzzy numbers:

$\tilde{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$, $\tilde{\mu} = (\mu_1, \mu_2, \mu_3)$.

For $\alpha \in [0,1]$, define their α-cuts by

$\tilde{\lambda}[\alpha] = [\lambda_1 + \alpha(\lambda_2 - \lambda_1),\ \lambda_3 - \alpha(\lambda_3 - \lambda_2)]$,

$\tilde{\mu}[\alpha] = [\mu_1 + \alpha(\mu_2 - \mu_1),\ \mu_3 - \alpha(\mu_3 - \mu_2)]$.

A sufficient stability condition for every α-level is

$\lambda U(\alpha) < \mu L(\alpha)$, for all $\alpha \in [0,1]$.

When this condition holds, the proposed interval performance measures are

$\tilde{L}[\alpha] = [\ \lambda L/(\mu U - \lambda L),\ \lambda U/(\mu L - \lambda U)\ ]$,

$\tilde{L}q[\alpha] = [\ \lambda L^2/\{\mu U(\mu U - \lambda L)\},\ \lambda U^2/\{\mu L(\mu L - \lambda U)\}\ ]$,

$\tilde{W}[\alpha] = [\ 1/(\mu U - \lambda L),\ 1/(\mu L - \lambda U)\ ]$,

$\tilde{W}q[\alpha] = [\ \lambda L/\{\mu U(\mu U - \lambda L)\},\ \lambda U/\{\mu L(\mu L - \lambda U)\}\ ]$.

3.3 Proposed Algorithm

- First of all Specify $\tilde{\lambda}$ and $\tilde{\mu}$ as triangular fuzzy numbers.
- Select α-levels in [0,1], e.g. 0, 0.1, …, 1.
- Compute the lower and upper endpoints of both α-cuts.
- Check $\lambda U(\alpha) < \mu L(\alpha)$. If violated, mark that α-level unstable.
- Evaluate L, Lq, W and Wq using the monotonic endpoint combinations.
- Repeat for all α-levels and store the resulting intervals.
- Compute the classical benchmark using the modal values λ_2 and μ_2.
- Compare crisp and fuzzy results and plot the uncertainty bands.

3.4 Numerical Example

Consider a single-server facility with uncertain arrival rate between 8 and 12 customers per hour, with 10 as the most plausible value, and uncertain service rate between 14 and 18 customers per hour, with 16 as the most plausible value:

$\tilde{\lambda} = (8, 10, 12)$, $\tilde{\mu} = (14, 16, 18)$.

Since $\lambda_3 = 12 < \mu_1 = 14$, the sufficient stability condition holds over the complete uncertainty range. At the modal values $\lambda = 10$ and $\mu = 16$, the classical M/M/1 results are L = 1.6667, Lq = 1.0417, W = 0.1667 hour, and Wq = 0.1042 hour.

3.5 α-Cut Numerical Results

α	λ interval	μ interval	L interval	Lq interval	W interval	Wq interval

0.0	[8.00, 12.00]	[14.00, 18.00]	[0.8000, 6.0000]	[0.3556, 5.1429]	[0.1000, 0.5000]	[0.0444, 0.4286]
0.2	[8.40, 11.60]	[14.40, 17.60]	[0.9130, 4.1429]	[0.4358, 3.3373]	[0.1087, 0.3571]	[0.0519, 0.2877]
0.4	[8.80, 11.20]	[14.80, 17.20]	[1.0476, 3.1111]	[0.5360, 2.3544]	[0.1190, 0.2778]	[0.0609, 0.2102]
0.6	[9.20, 10.80]	[15.20, 16.80]	[1.2105, 2.4545]	[0.6629, 1.7440]	[0.1316, 0.2273]	[0.0721, 0.1615]
0.8	[9.60, 10.40]	[15.60, 16.40]	[1.4118, 2.0000]	[0.8264, 1.3333]	[0.1471, 0.1923]	[0.0861, 0.1282]
1.0	[10.00, 10.00]	[16.00, 16.00]	[1.6667, 1.6667]	[1.0417, 1.0417]	[0.1667, 0.1667]	[0.1042, 0.1042]

Table 1. Illustrative α-cut intervals generated by the proposed fuzzy M/M/1 model.

3.6 Comparison with Previous Models

Study/model	Method	Systems	Output	Main feature
Li & Lee (1989)	Extension principle, possibility, fuzzy Markov chains	M/F/1, FM/FM/1	Analytical fuzzy measures	Early general analytical framework
Negi & Lee (1992)	α-cuts and simulation	Fuzzy queues with trapezoidal numbers	Analytical + simulation	Combines fuzzy and probabilistic approaches
Kao, Li & Chen (1999)	α-cuts + parametric programming	M/F/1, F/M/1, F/F/1, FM/FM/1	Membership functions	Systematic construction of performance memberships

Proposed model	α-cut interval propagation + stability check	Fuzzy M/M/1	L, Lq, W, Wq intervals	Simple reproducible algorithm and direct crisp comparison

Table 2. Literature-based comparison. The descriptions of the earlier models are based on the cited publications.

3.7 Crisp versus Fuzzy Comparison

Measure	Classical modal	Fuzzy α=0	Fuzzy α=0.5	Fuzzy α=1
L	1.6667	[0.8000, 6.0000]	[1.1250, 2.7500]	[1.6667, 1.6667]
Lq	1.0417	[0.3556, 5.1429]	[0.5956, 2.0167]	[1.0417, 1.0417]
W	0.1667	[0.1000, 0.5000]	[0.1250, 0.2500]	[0.1667, 0.1667]
Wq	0.1042	[0.0444, 0.4286]	[0.0662, 0.1833]	[0.1042, 0.1042]

Table 3. Comparison of the classical modal result with selected fuzzy α-cut intervals.

3.8 Graphical Interpretation

Figure 1 illustrates the contraction of the uncertainty interval for the mean number of customers in the system as α increases. At α = 0 the full specified uncertainty is retained. At α = 1 the fuzzy parameters reduce to their modal values and the interval collapses to the classical result.

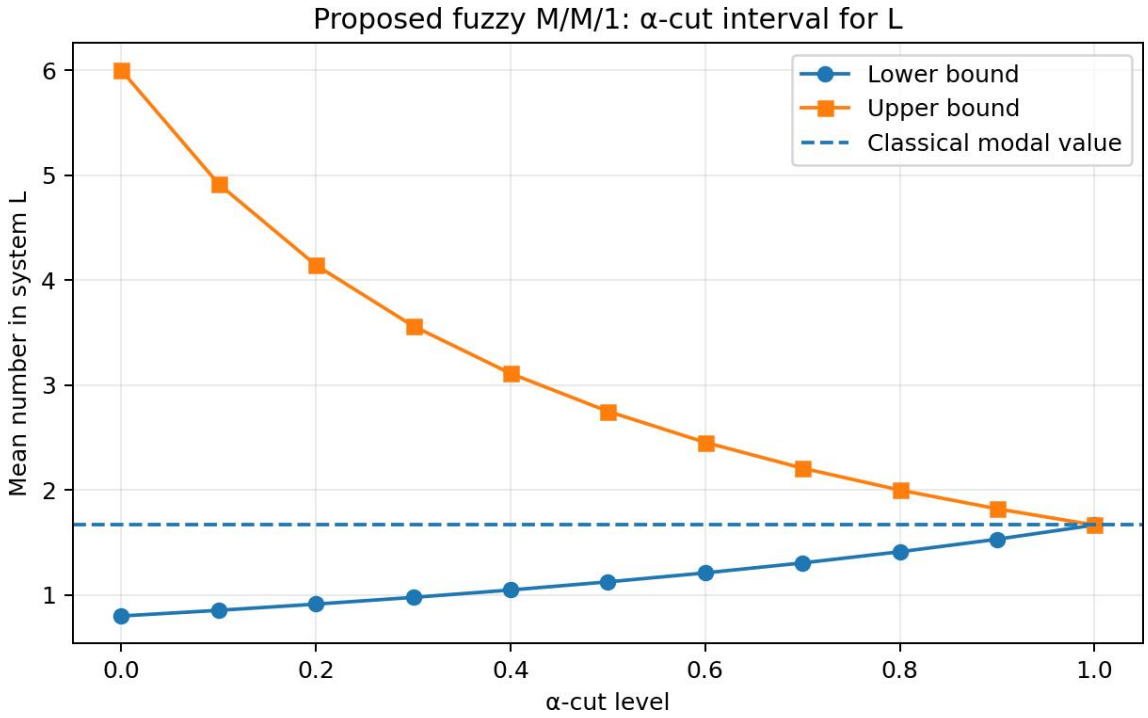


Figure 1. α-cut uncertainty interval for L in the proposed fuzzy M/M/1 model.

3.9 Importance of the Proposed Framework

The numerical example shows that the proposed model gives more information than a single crisp estimate. A decision-maker can inspect conservative, intermediate and most-plausible operating scenarios. The method is computationally simple, transparent, and directly implementable in MATLAB, Python or Excel.

The proposed framework complements rather than replaces previous fuzzy queueing methods. Li and Lee emphasize extension-principle and fuzzy Markov-chain analysis; Negi and Lee combine fuzzy modeling with simulation; and Kao, Li and Chen use α-cut parametric programming to derive membership functions. The present model focuses on direct interval performance bands for a standard M/M/1 queue and can serve as a baseline for extensions to M/M/c, finite-capacity and priority systems.

4. Theoretical Results

Let the fuzzy arrival and service rates be represented by triangular fuzzy numbers

$\tilde{\lambda} = (\lambda_1, \lambda_2, \lambda_3)$, $\tilde{\mu} = (\mu_1, \mu_2, \mu_3)$,

where $\lambda_1 \le \lambda_2 \le \lambda_3$ and $\mu_1 \le \mu_2 \le \mu_3$. For $\alpha \in [0,1]$, define

$$\lambda L(\alpha) = \lambda_1 + \alpha(\lambda_2 - \lambda_1),\ \lambda U(\alpha) = \lambda_3 - \alpha(\lambda_3 - \lambda_2),$$
$$\mu L(\alpha) = \mu_1 + \alpha(\mu_2 - \mu_1),\ \mu U(\alpha) = \mu_3 - \alpha(\mu_3 - \mu_2).$$

Suppose c identical servers are available. Define the worst-case utilization at level α by

$\rho U(\alpha) = \lambda U(\alpha) / [c\, \mu L(\alpha)]$.

We introduce the following robustness index:

$R(\alpha) = 1 - \lambda U(\alpha) / [c\, \mu L(\alpha)]$,

whenever $\lambda U(\alpha) < c\, \mu L(\alpha)$.

Theorem 4.1. α-Level Stability Theorem

For the proposed fuzzy M/M/c queue, if

$\lambda U(\alpha) < c\, \mu L(\alpha)$, for all $\alpha \in [0,1]$,

then the queueing system is stable for every α-level. Consequently, $0 < \rho U(\alpha) < 1$ and $R(\alpha) > 0$ for all $\alpha \in [0,1]$.

Proof

For a fixed $\alpha \in [0,1]$, the possible arrival rate belongs to $[\lambda L(\alpha), \lambda U(\alpha)]$ and the possible service rate of each server belongs to $[\mu L(\alpha), \mu U(\alpha)]$. Therefore, the largest possible arrival rate is $\lambda U(\alpha)$, whereas the smallest possible service capacity of c servers is $c\mu L(\alpha)$.

Hence the worst-case traffic intensity is

$\rho U(\alpha) = \lambda U(\alpha) / [c\, \mu L(\alpha)]$.

By assumption, $\lambda U(\alpha) < c\mu L(\alpha)$. Dividing by the positive quantity $c\mu L(\alpha)$, we obtain $\rho U(\alpha) < 1$. Since the arrival and service rates are positive, $\rho U(\alpha) > 0$. Thus $0 < \rho U(\alpha) < 1$.

The classical stability condition for an M/M/c queue is $\rho < 1$. Hence the queue is stable at the selected α-level. Since the argument holds for every $\alpha \in [0,1]$, the proposed fuzzy queue is stable throughout the complete fuzzy uncertainty domain.

Finally, $R(\alpha) = 1 - \rho U(\alpha)$. Since $0 < \rho U(\alpha) < 1$, it follows that $R(\alpha) > 0$. Therefore, the proposed system is robustly stable over the entire α-domain.

Theorem 4.2. Monotonicity of the Robustness Index

For a fixed number of servers c, the proposed robustness index

$R(\alpha) = 1 - \lambda U(\alpha) / [c\, \mu L(\alpha)]$

is non-decreasing with respect to α, provided $\lambda_3 < c\mu_1$. In other words, increasing the confidence level α cannot decrease the robustness of the system.

Proof

We have

$\lambda U(\alpha) = \lambda_3 - \alpha(\lambda_3 - \lambda_2)$,

$d\lambda U/d\alpha = -(\lambda_3 - \lambda_2) \le 0.$

Similarly,

$\mu L(\alpha) = \mu_1 + \alpha(\mu_2 - \mu_1),$

$d\mu L/d\alpha = \mu_2 - \mu_1 \ge 0.$

Define

$q(\alpha) = \lambda U(\alpha) / [c\, \mu L(\alpha)].$

Differentiating gives

$q'(\alpha) = (1/c)\, [\lambda U'(\alpha)\, \mu L(\alpha) - \lambda U(\alpha)\, \mu L'(\alpha)] / \mu L^2(\alpha).$

Because $\lambda U'(\alpha) \le 0$, $\mu L'(\alpha) \ge 0$, and $\lambda U(\alpha)$, $\mu L(\alpha)$ are positive, both terms in the numerator are non-positive. Hence $q'(\alpha) \le 0$.

Therefore, $\lambda U(\alpha)/[c\mu L(\alpha)]$ is non-increasing in α. Since $R(\alpha) = 1 - q(\alpha)$,

$R'(\alpha) = -q'(\alpha) \ge 0.$

Hence $R(\alpha)$ is non-decreasing on $[0,1]$.

Interpretation

At $\alpha = 0$, the model considers the complete uncertainty range and therefore provides the most conservative robustness assessment. At $\alpha = 1$, only the modal values remain, so $\lambda U(1) = \lambda_2$ and $\mu L(1) = \mu_2$. Consequently,

$$R(1) = 1 - \lambda_2/(c\, \mu_2).$$

Thus, the proposed robustness index connects the worst-case fuzzy scenario with the classical modal scenario.

Theorem 4.3. Optimal Server Allocation Theorem

Suppose that a decision-maker requires the fuzzy queueing system to maintain a minimum robustness level $r_0 \in (0,1)$. Define the required number of servers at confidence level α by

$c^*(\alpha) = \min \{ c \in N : R(\alpha) \ge r_0 \}.$

Then

$c^*(\alpha) = \lceil \lambda U(\alpha) / [(1 - r_0)\, \mu L(\alpha)] \rceil .$

Proof

The required robustness condition is $R(\alpha) \ge r_0$. Using the definition of $R(\alpha)$,

$1 - \lambda U(\alpha)/[c\, \mu L(\alpha)] \ge r_0 .$

Rearranging,

$1 - r_0 \ge \lambda U(\alpha)/[c\, \mu L(\alpha)].$

Since $1 - r_0 > 0$,

$c(1 - r_0)\, \mu L(\alpha) \ge \lambda U(\alpha).$

Consequently,

$c \ge \lambda U(\alpha) / [(1 - r_0)\, \mu L(\alpha)].$

Because the number of servers must be a positive integer, the smallest feasible value is obtained by taking the ceiling:

$c^*(\alpha) = \lceil \lambda U(\alpha) / [(1 - r_0)\, \mu L(\alpha)] \rceil .$

Therefore, $c^*(\alpha)$ is the minimum number of servers required to achieve the prescribed robustness level r_0 at confidence level α.

Numerical Illustration

Consider

$\tilde{\lambda} = (18, 22, 27), \quad \tilde{\mu} = (7, 8, 9).$

Suppose the required robustness level is $r_0 = 0.20$.

At $\alpha = 0$, $\lambda U(0) = 27$ and $\mu L(0) = 7$.

Therefore,

$c^*(0) = \lceil 27 / [(1 - 0.20)(7)] \rceil = \lceil 27/5.6 \rceil = 5.$

Hence five servers are required to guarantee at least 20% robustness under the full uncertainty range.

At $\alpha = 1$, $\lambda U(1) = 22$ and $\mu L(1) = 8$. Therefore,

$c^*(1) = \lceil 22 / [(1 - 0.20)(8)] \rceil = \lceil 22/6.4 \rceil = 4.$

Thus, under the modal scenario, four servers are required.

Interpretation

The result demonstrates an important managerial implication: four servers may be adequate under the most plausible operating condition, whereas five servers are required for robust protection against the complete uncertainty range.

5. Application of the Proposed Model to Hospital Emergency Department Staffing

The proposed Robustness and Adaptive Server Allocation framework can be applied to hospital emergency departments, where patient arrivals and treatment capacities are inherently uncertain. Patients do not arrive at a constant rate, and the treatment time of

each patient varies according to the severity and complexity of the case. Consequently, both the arrival rate and the service rate can be represented by fuzzy numbers rather than fixed numerical quantities.

5.1 Description of the Emergency Department

Consider a hospital emergency department in which doctors or treatment stations operate in parallel. Patients arriving at the emergency department form the customer population, while doctors or treatment stations act as servers. The waiting room represents the queue. The principal objective is to determine the minimum number of active doctors required to maintain a prescribed level of robustness against uncertain patient demand and uncertain treatment capacity.

Let the fuzzy patient arrival rate be represented by $\tilde{\lambda} = (18, 22, 27)$ patients/hour, where 18 patients/hour represents the lower estimate, 22 patients/hour is the most plausible arrival rate, and 27 patients/hour represents the upper estimate.

Similarly, let the fuzzy service rate of an individual doctor be $\tilde{\mu} = (7, 8, 9)$ patients/hour/doctor, where 7 patients/hour is the lower service capacity, 8 patients/hour is the most plausible service rate, and 9 patients/hour is the upper service capacity.

Suppose that the hospital requires a minimum robustness level of $r_0 = 0.20$.

5.2 Robustness-Based Staffing Model

The proposed robustness index for c active doctors is

$R(\alpha) = 1 - \lambda U(\alpha) / [c\, \mu L(\alpha)]$.

The minimum number of doctors required at confidence level α is obtained from Theorem 3 as

$c^*(\alpha) = \lceil \lambda U(\alpha) / [(1 - r_0)\, \mu L(\alpha)] \rceil$.

The hospital therefore does not need to select the staffing level only from a single estimated arrival rate. Instead, it can determine the required number of doctors by considering the uncertainty interval at each α-level.

5.3 Worst-Case Staffing Requirement: α = 0

At α = 0, the complete uncertainty range is considered. Therefore,

$\lambda U(0) = 27, \quad \mu L(0) = 7.$

Using the prescribed robustness level $r_0 = 0.20$, Theorem 4.3 gives

$$c^*(0) = \lceil 27 / [(1 - 0.20)(7)] \rceil . \; c^*(0) = \lceil 27 / 5.6 \rceil = \lceil 4.8214 \rceil = 5.$$

Hence, five doctors or treatment servers are required to guarantee at least 20% robustness under the complete uncertainty range.

5.4 Most Plausible Staffing Requirement: α = 1

At α = 1, only the modal values of the fuzzy parameters are considered. Thus,

$\lambda U(1) = 22, \quad \mu L(1) = 8.$

Consequently,

$$c^*(1) = \lceil 22 / [(1 - 0.20)(8)] \rceil . \; c^*(1) = \lceil 22 / 6.4 \rceil = \lceil 3.4375 \rceil = 4.$$

Therefore, four doctors are sufficient under the most plausible operating condition.

5.5 Staffing Requirement at Different α-Levels

The following table demonstrates how the required number of doctors changes as the confidence level α increases.

α-level	λU(α)	μL(α)	Required doctors c*(α)	Interpretation

0.0	27.00	7.00	5	Complete uncertainty
0.2	26.00	7.20	5	High uncertainty
0.4	25.00	7.40	5	Moderate-high uncertainty
0.6	24.00	7.60	4	Moderate uncertainty
0.8	23.00	7.80	4	Low uncertainty
1.0	22.00	8.00	4	Most plausible condition

Table 4. Adaptive staffing requirement at different α-levels.

5.6 Verification of the Robustness Requirement

Suppose the hospital deploys four doctors. Under the complete uncertainty scenario $\alpha = 0$, the robustness index is

$R(0) = 1 - 27/[4(7)]$.

$R(0) = 1 - 27/28 = 0.0357$.

Thus, the robustness level is only approximately 3.57%, which is substantially lower than the required 20%. Therefore, four doctors are not sufficient for robust operation under the worst-case uncertainty scenario.

Now consider five doctors. Then

$R(0) = 1 - 27/[5(7)]$.

$R(0) = 1 - 27/35 = 0.2286$.

Hence, the robustness level is approximately 22.86%, which exceeds the required 20%. Therefore, five doctors satisfy the hospital's robustness requirement.

5.7 Operational Interpretation

The proposed model provides hospital administrators with a direct staffing decision rule. If the objective is to protect the emergency department against the full uncertainty range, five doctors should be available. If the hospital is operating under a normal condition corresponding to a high α-level, four doctors may be adequate.

The distinction between these two staffing decisions is important. A conventional model based only on the most likely values $\lambda = 22$ and $\mu = 8$ would recommend four doctors. However, such a decision may provide insufficient protection when the actual arrival rate approaches 27 patients/hour and the service rate falls toward 7 patients/hour. The proposed fuzzy model explicitly captures this risk.

5.8 Practical Staffing Policy

Based on the proposed model, the emergency department can adopt the following staffing policy:

- During normal operating conditions, four doctors may be deployed when the uncertainty level is relatively low.
- When patient demand becomes more uncertain, the system should increase the active staffing level.
- For the complete uncertainty range considered in this example, five doctors are required to maintain at least 20% robustness.
- The robustness index can be recalculated whenever new estimates of patient arrival and treatment rates become available.
- The same framework can be extended to nurses, treatment rooms, ambulances, ICU beds, or other parallel service resources.

5.9 Managerial Significance

The application demonstrates that the proposed fuzzy queueing framework can transform uncertain operational information into an explicit staffing decision. Instead of reporting only an expected queue length or waiting time, the model answers the practical question: how many service providers should be deployed to maintain a specified robustness level under uncertain demand and service capacity?

For the present emergency-department example, the model identifies a clear difference between nominal and robust staffing: four doctors are sufficient at the most plausible α = 1 condition, whereas five doctors are required when the complete uncertainty range is considered. This provides a quantitative basis for contingency staffing during periods of elevated or uncertain emergency demand.

5.10 Generalization to Other Healthcare Systems

The same application framework can be generalized to other healthcare queueing systems. For example, patients may be treated as customers and doctors, nurses, diagnostic machines, operating rooms, or ambulances may be treated as servers. The fuzzy arrival and service parameters can be estimated from historical records, expert opinion, or linguistic assessments such as low, normal, high, and very high demand.

Thus, the proposed model can support robust resource allocation in emergency departments, outpatient clinics, diagnostic centres, ambulance dispatch systems, ICU admission systems, and hospital pharmacy services.

Observation

The hospital emergency-department application illustrates the practical value of the proposed theoretical results. The α-level stability theorem guarantees a stable operating region, the monotonicity theorem explains the behavior of the robustness index as uncertainty is reduced, and the optimal server allocation theorem converts the required robustness into a concrete staffing decision. Hence, the framework establishes a direct connection between fuzzy mathematical modelling and real-world resource allocation.

6. Discussion and Limitations

The numerical results show why a fuzzy treatment can lead to a different operational interpretation from a single crisp calculation. In the M/M/1 example, the modal values λ=10 and μ=16 give L=1.6667. When the complete support of the fuzzy parameters is considered, however, L lies in [0.8000, 6.0000]. The upper end reflects the combination of relatively high arrivals and relatively low service capacity. As α increases, both parameter intervals contract and the performance interval moves towards the modal calculation.

The multi-server example gives a similar interpretation from a staffing perspective. With $\tilde{\lambda}$ = (18, 22, 27), $\tilde{\mu}$ = (7, 8, 9), and a target robustness of 0.20, four servers satisfy the criterion at α=1 but not at α=0. At α=0, four servers give R(0)=0.0357, whereas five servers give R(0)=0.2286. The extra server therefore acts as a protection margin against the simultaneous occurrence of high demand and low service capacity.

The formulation also has limitations. Triangular membership functions are assumed for both arrival and service rates. This is convenient when lower, modal, and upper estimates are available, but it may not represent asymmetric or multimodal assessments. In addition, the robustness calculation deliberately uses the conservative combination λU(α) and μL(α); it is therefore a worst-case screening measure rather than a probabilistic estimate of how frequently that combination occurs. Finally, the hospital example is illustrative, and its parameter values should not be interpreted as measurements from a particular emergency department.

For an empirical implementation, the fuzzy parameters should be estimated from local hospital records, expert elicitation, or a documented combination of both. These limitations point to natural directions for validation rather than changing the purpose of the present framework.

Conclusion

The analysis presented here shows how uncertainty in arrival and service rates can be carried through standard queueing formulae without first replacing the uncertain inputs by single point estimates. For the fuzzy M/M/1 example, the α-cut intervals for L, Lq, W, and Wq become progressively narrower as α increases and collapse to the classical modal values at α = 1. The stability test λU(α) < μL(α) also gives a direct way of identifying the α-levels for which the selected queue remains stable.

The extension to M/M/c introduces a robustness measure based on worst-case utilization. The resulting theorems show that robustness is non-decreasing with α and give an explicit integer formula for the minimum number of servers needed to achieve a prescribed robustness level. In the staffing example, four servers are sufficient at the modal condition, whereas five are needed when the full uncertainty range is used with a 20% robustness target.

The hospital example is intended to demonstrate the decision mechanism rather than to claim a universal staffing rule. In an actual emergency department, the three-point fuzzy parameters should be obtained from local data and expert assessment. The example nevertheless makes the main practical benefit clear: the uncertainty buffer is made explicit instead of being hidden inside a single estimated arrival or service rate.

A useful next step is to test the framework with empirical hospital data and compare its staffing recommendations with simulation and classical stochastic queueing models. Other extensions include heterogeneous servers, finite-capacity and priority queues, time-varying rates, retrial systems, and hybrid fuzzy-probabilistic formulations.

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